

DA2 Momentum Resolution of tracking Detector

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https://github.com/Leon0605/DA2_Momentum-Resolution-of-tracking-Detector

Abstract

This paper discusses the simulation and evaluation of a simple particle tracking detector to track the position of a charged particle fired through several successive, evenly-spaced detection planes. From this, we estimate both the point of creation as well as the tracking and momentum resolutions of the particle's trajectory. The experiment is thus split into two parts, the first of which concerns itself with only five detection planes. The second part introduces a magnetic field to influence the particle's trajectory, after which three more detection planes measure the change in trajectory. Using Monte Carlo methods and least-squares fits, we estimate the point of creation as well as the tracking and momentum resolutions of the particle, varying values for the true momentum and the magnet strength. Altering the magnet strength decreases the uncertainty of the resolution, while increasing the momentum increases the uncertainty and the mean. We hypothesize that the improvement with stronger magnets is due to the larger trajectory deviation, while the degradation with higher momentum results from the lesser deflection by the magnet, which, combined with the inherent uncertainty of the hit positions, compromises the reconstruction. We further identify two sources of systematic bias, hit-position discretizations, and the small-angle approximation in the momentum reconstruction, and show that adopting an exact geometric expression for the bending angle effectively eliminates the latter.

Introduction

In this experiment, we will estimate both the point of creation and the momentum of charged particles by building and simulating a particle tracking detector to reconstruct their respective trajectories. To estimate the momentum, the particle will be passed through a homogeneous magnetic field, flanked by vertical, equally-spaced detection planes on either side.

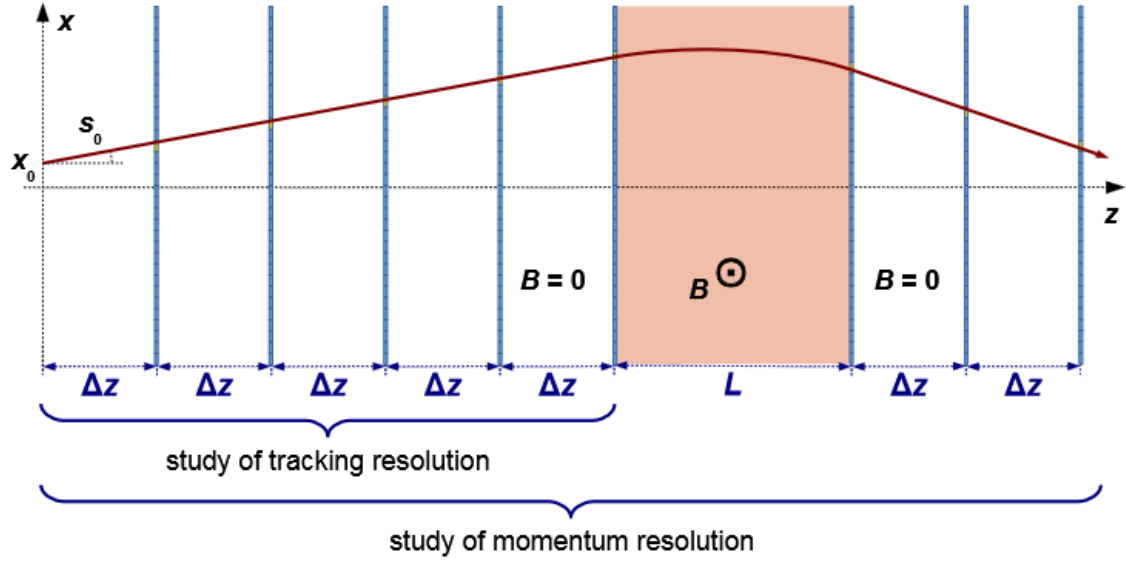


Figure 1: *Graphical illustration of the experimental setup.*

x_0 denotes the starting point of the particle on the x -axis, s_0 the angle of the particle's trajectory, and B the strength of the magnetic field, expressed in T . Δz denotes the space between each detector plane along the z -axis, and L the length of the magnetic field. Each detection plane is made up of cells with $\text{cellWidth} = 500\mu\text{m}$ that detect when a particle passes through them. There are five and three detection planes in front and behind the magnetic field, respectively.

Methodology

The experiment consists of two parts. We estimate the tracking resolution in the first part, and the momentum resolution in the second. We will use the pull to check our model for possible biases or wrong uncertainty estimations:

$$\text{pull} = \frac{\text{reconstructed quantity} - \text{generated quantity}}{\text{uncertainty on reconstructed quantity}} \quad (1)$$

Tracking Resolution

For this first part, we only concern ourselves with the first n detection planes in front of the magnet, such that in this first setup, the trajectory of the particle does not change. We will simulate 1'000 trajectories, defining $n = 5$, $\Delta z = 2 \text{ cm}$, $z_0 = 0$, $x_0^{\text{true}} = \mathcal{N}(0, 1 \text{ mm})$, and $s_0^{\text{true}} = \tan(\mathcal{N}(0, 0.1 \text{ rad}))$, where $s_0 = \tan(\alpha)$ denotes the slope of the trajectory, with α being the angle with respect to the z -axis. For each detector response of the n detection planes, we determine which cell was hit. This is done using:

$$\text{cellHit} = \left\lfloor \frac{x}{\text{cellWidth}} \right\rfloor + 1, \quad x \geq 0 \quad (2)$$

$$\text{cellHit} = \left\lfloor \frac{x}{\text{cellWidth}} \right\rfloor, \quad x < 0 \quad (3)$$

This ensures that the ID of the cells hit is 1 or greater for $x \geq 0$ and -1 or smaller for $x < 0$, such that there is no +0 and -0 cell ID of the hit cell. Then this is used to reconstruct the x-position by taking the center of the hit cell, with an uncertainty of $\sigma = \frac{\text{cellWidth}}{\sqrt{12}}$, corresponding to the standard deviation of a uniform distribution over the cell width. The reconstructed hit positions are then passed into a least-squares fitter to estimate the trajectory, modeled as a straight line, as no magnetic field is present, returning the starting position x_0 and the slope s_0 . These reconstructed parameters are then used to calculate pull distributions, which evaluate whether the reconstruction is unbiased and the uncertainties are correctly estimated.

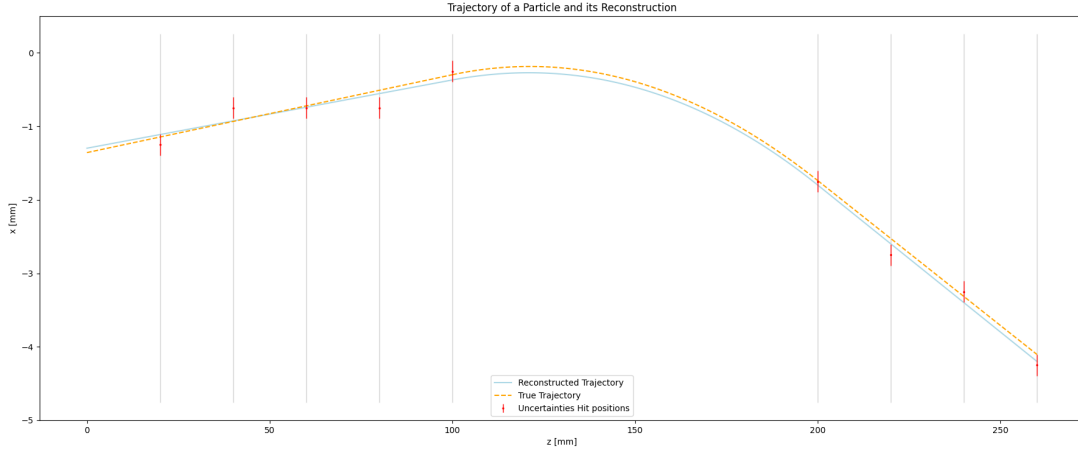


Figure 2: *Plot of true trajectory and reconstructed trajectory with uncertainties on hit points through the detection planes of the detector.*

Momentum Resolution

For this second part, we will take into account all eight detection panels, as well as the magnetic field. To calculate the change in trajectory of the particle as it passes through the magnet field, we will first determine the radius of curvature ρ

$$\rho = \frac{p_T}{q \cdot B}, \quad (4)$$

expressed in meters, while q is charge of the particle, in units of elementary charge; B is the strength of the magnetic field, in T; and

$$p_T \approx 0.3 \cdot q \cdot B \cdot \rho \quad (5)$$

is the momentum transverse to the magnetic field lines in GeV. We define $n = 5$, $\Delta z = 2$ cm, $z_0 = 0$, $L = 10$ cm, $B = 0.5$ T, $x_0^{true} = \mathcal{N}(0, 1 \text{ mm})$, $s_0^{true} = \tan(\mathcal{N}(0, 0.1 \text{ rad}))$, and $p_T^{true} = 0.3$ GeV. For q , we randomly assign a charge $+1$ or -1 . Once again, we will generate 1'000 particles.

We will run the simulation for $p_T^{true} = \{0.1, 0.3, 1, 2, 5, 10, 20\}$ GeV and $B = 0.5$ T, as well as $p_T^{true} = 0.3$ GeV and $B = \{1.0, 1.5, 2.0\}$ T.

Under the assumption that the radius of the particle trajectory is much larger than the length of the magnetic field, we estimate the uncertainty of p_T^{reco} using the bending angle

$$\theta \approx \frac{L}{\rho} = \frac{L \cdot q \cdot B}{p_T}, \quad (6)$$

rephrasing it such that:

$$p_T^{reco} \approx \frac{L \cdot q \cdot B}{\theta} \Leftrightarrow p_T^{reco} \approx \frac{L \cdot q \cdot B}{|\arctan(s_0^{reco}) - \arctan(s_1^{reco})|} \quad (7)$$

However, since this formula introduced a systematic bias, we instead derived an exact analytical formula

$$p_T = \frac{L \cdot q \cdot B}{\sin(\theta_{\text{out}}) - \sin(\theta_{\text{in}})}, \quad (8)$$

where $\theta_i = \arctan(s_i)$. With this formula the systematic bias was eliminated because it does not rely on the small-angle approximation, which breaks down when the radius is not much larger than the magnet length. More information on the derivation is given in the Problems section.

To obtain the uncertainty of p_T from the uncertainties of the input slope s_0 and the output slope s_1 , we used error propagation to derive the following two formulas. For the first approach to determine the reconstruction of p_T we got

$$\sigma(p_T) = |q \cdot B \cdot L| \cdot \left(\frac{\sigma_{s_0}^2}{(1 + s_0^2)^2 \cdot |\Delta(s_0, s_1)|^4} + \frac{\sigma_{s_1}^2}{(1 + s_1^2)^2 \cdot |\Delta(s_0, s_1)|^4} \right)^{\frac{1}{2}} \quad (9)$$

and for the new exact formula

$$\sigma(p_T) = |q \cdot B \cdot L| \cdot \frac{\sqrt{(s_0^2 + 1)^3 \sigma_{s_1}^2 + (s_1^2 + 1)^3 \cdot \sigma_{s_0}^2}}{\sqrt{s_1^2 + 1} \sqrt{s_0^2 + 1} \left(s_0 \sqrt{s_1^2 + 1} - s_1 \sqrt{s_0^2 + 1} \right)^2} \quad (10)$$

Further details are elaborated on in the Appendix, subsection B. Where not explicitly mentioned otherwise, all formulas were derived using SI units; in actual calculations, we used gigaelectronvolt (GeV) for p_T , simply multiplying the respective equation by 0.3.

Results

In this section we talk about our findings of using our methods on the generated particle data.

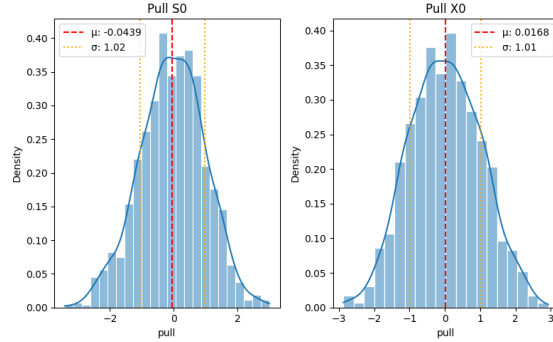


Figure 3: *Histograms showing the pull distribution over 1'000 repetitions for the tracking resolution*

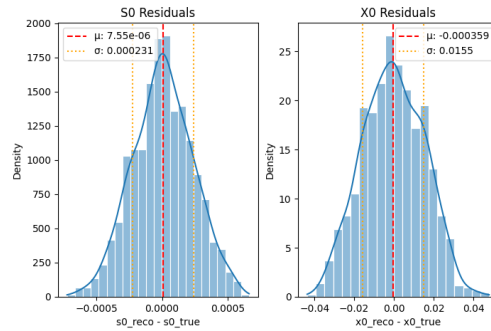


Figure 4: *Histograms showing the residuals distribution over 1'000 repetitions for the tracking resolution*

From figure 3 we can observe that the pull distribution is nearly a standard normal distribution which indicates a lack of bias and correct modeling of the uncertainties. For the resolution, the residuals are of magnitude 10^{-4} to 10^{-6} with low uncertainties of 0.00023 for the slope s_0 and 0.016 for the starting position x_0 , which, accounting for the uncertainty, is consistent with a mean of zero.

Using a Student's t-test and a χ^2 -test with $\alpha = 0.0001\%$, we checked whether the mean and standard deviations of our estimated p_T , s_0 , and x_0 values deviated significantly from 0 and 1, respectively. We ran the experiment and the corresponding hypothesis tests 100 times in order to get a good idea of how well it performs over several repetitions, therefore, several datasets, as each run is subject to its own stochastic uncertainty.

Table 1 shows the rejection rates,

Hypothesis	Variable	Rejections (out of 100)
$H_0(\mu = 0)$	p_T	0
$H_0(\mu = 0)$	s_0	0
$H_0(\mu = 0)$	x_0	0
$H_0(\sigma = 1)$	p_T	0
$H_0(\sigma = 1)$	s_0	34
$H_0(\sigma = 1)$	x_0	0

Table 1: *Rejection rates ($\alpha = 0.0001\%$), using a Student's t-test for the mean and a χ^2 -test for the standard deviation.*

which are zero for all variables except for σ of s_0 , where 34 of 100 runs failed the test. This indicates that the uncertainty model is correct for p_T and x_0 , but underestimates the uncertainty of s_0 , a consequence of the non-Gaussian (uniform) measurement errors further discussed in the Problems section. As described in more detail in the problems section, we encountered a systematic bias caused by the approximate formula $\theta \approx \frac{L}{\rho} = \frac{L \cdot q \cdot B}{p_T}$, which was eliminated by calculating the exact p_T .

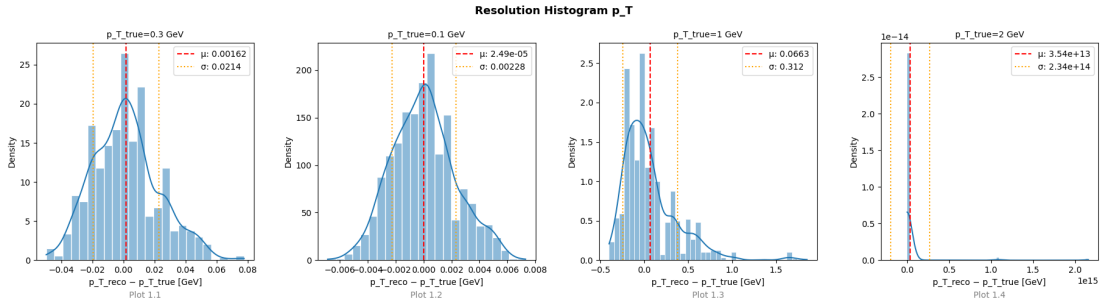


Figure 5: *Histograms showing the distribution of the difference $p_T^{\text{reco}} - p_T^{\text{true}}$ over 1'000 repetitions for different values of $p_T^{\text{true}} = \{0.1, 0.3, 1, 2\} \text{ GeV}$, with $B = 0.5 \text{ T}$, including the respective means and standard deviations.*

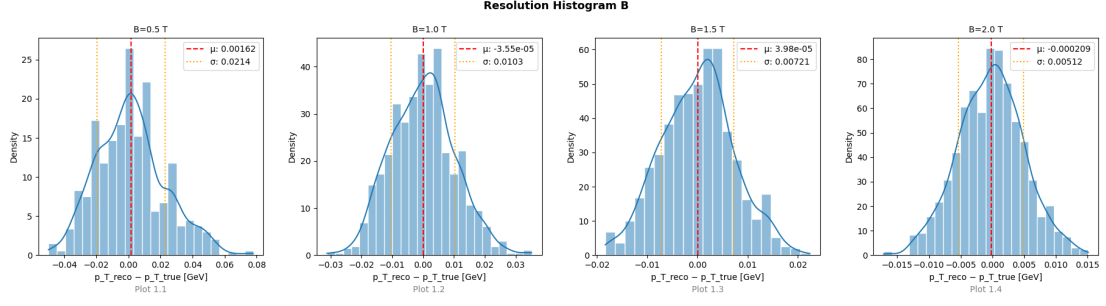


Figure 6: Histograms showing the distribution of the difference $p_T^{\text{reco}} - p_T^{\text{true}}$ over 1'000 repetitions for different values of $B = \{0.5, 1.0, 1.5, 2.0\}$ T, with $p_T = 0.3$ GeV, including the respective means and standard deviations.

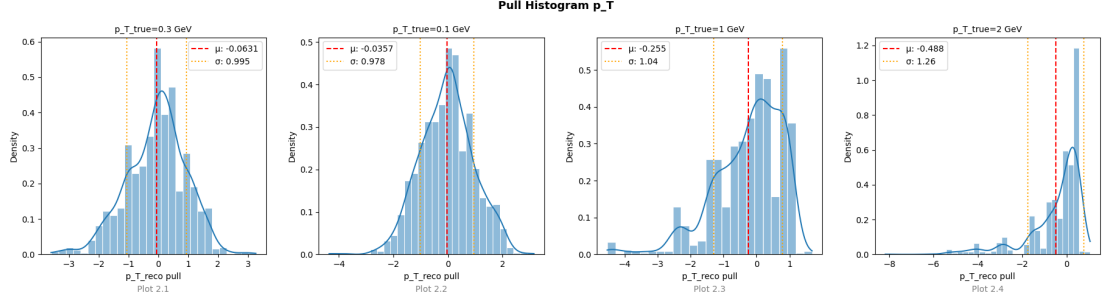


Figure 7: Histograms showing the pull distribution over 1'000 repetitions for different values of $p_T^{\text{true}} = \{0.1, 0.3, 1, 2\}$ GeV, with $B = 0.5$ T, including the respective means and standard deviations.

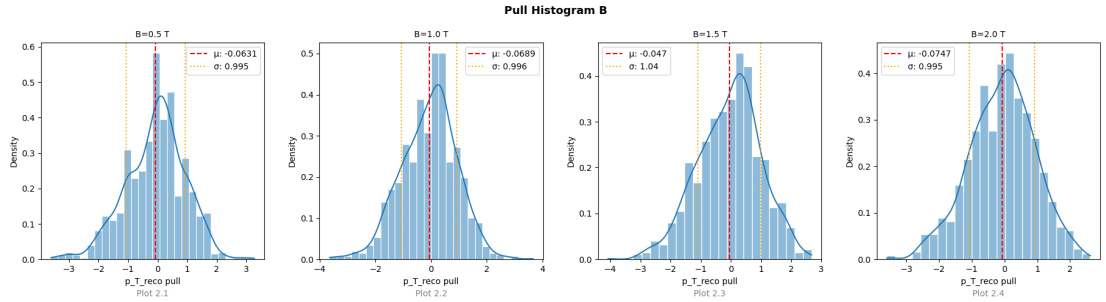


Figure 8: Histograms showing the pull distribution over 1'000 repetitions for different values of $B = \{0.5, 1.0, 1.5, 2.0\}$ T, with $p_T = 0.3$ GeV, including the respective means and standard deviations.

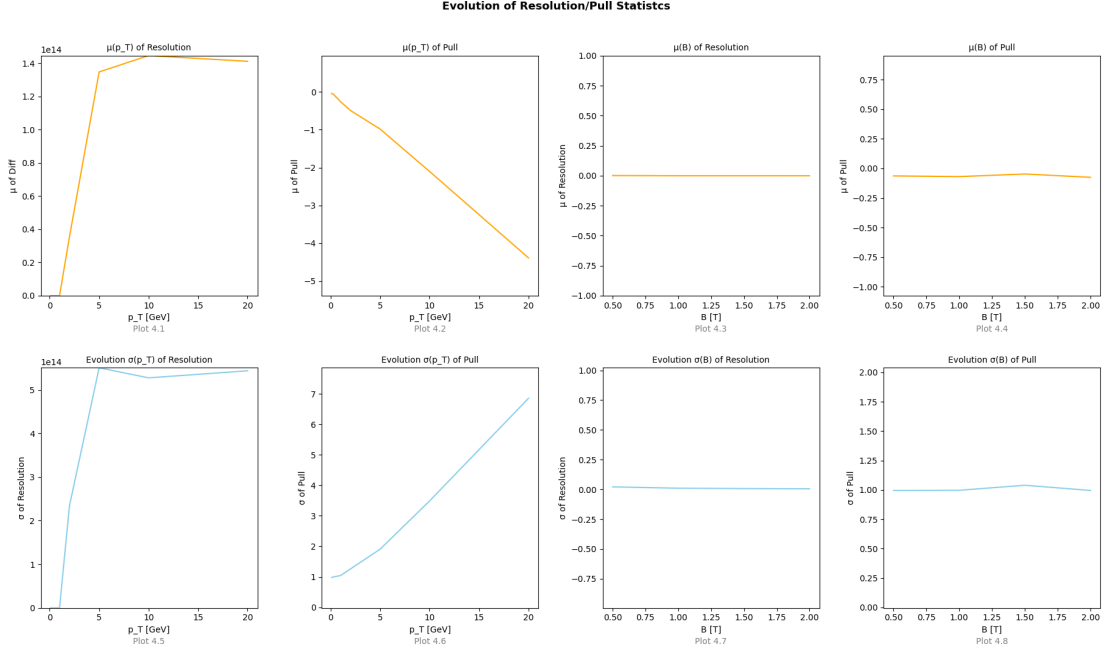


Figure 9: Plots showing the changes in the mean (μ) and standard deviation (σ) of the resolution and pull distributions over different values of p_T^{true} ($\{0.1, 0.3, 1, 2, 5, 10, 20\}$ GeV, with $B = 0.5$ T) and B ($\{1.0, 1.5, 2.0\}$ T, with $p_T = 0.3$ GeV).

From the next plots we just show the first four where the trend can be best discerned from, and the plots with $p_T > 2$ GeV do not give much more information. They can be viewed in the Appendix, subsection A (fig. 12–16). From figures 5, 7, and 9, we can discern that for increasing values of p_T , both the mean and standard deviation of the resolution increases drastically, plateauing at $p_T = 5$ GeV. Similarly, the mean of the pull drops from 0 rather linearly with increasing p_T ; interestingly, an opposite trend is observed for the standard deviation. Altering B has almost no effect on the mean but as observable in Fig. 6, the standard deviation of the resolution decreases, which indicates a more stable reconstruction with increased magnet strength. For the mean and standard deviation of the pull we observe in Fig. 8 that there is no critical change, showing us a rather good simulation even for increased magnet strength. Possible explanations for the findings with regard to the behaviour observed when changing p_T could be that a higher p_T (which is the product of both the mass and velocity of the particle) leads to a larger bending radius $\rho = p_T/(q \cdot B)$, and therefore a smaller deflection angle $\theta \approx L/\rho$, or in the new formula $\sin(\theta_{\text{out}}) - \sin(\theta_{\text{in}})$. This would have resulted in less variability between repetitions, and the uncertainty brought on by the detector resolution could have thus caused the increased mean and variance of the resolution as it failed to pick up on minor deviations. Furthermore, since the deflection angle appears in the denominator of the reconstruction formula, small angles lead to an explosive behaviour of the resolution.

We conclude that increasing the magnet strength improves the momentum reconstruction. However, the resolution still significantly degrades for particles with higher momentum, even with stronger magnetic fields.

Problems

In this section we discuss the biggest problems encountered while working on the simulation of the particle detector. There were three major problems we needed to solve.

The first problem we encountered concerned the estimation of hit positions. Hit positions were estimated by placing each hit at the center of its fired cell, with uncertainty $\sigma = \frac{\text{cellWidth}}{\sqrt{12}}$. The estimated hit points were then passed into a least-squares fit with the uncertainty as absolute weights. When evaluating pull distributions, the pull width for the reconstructed slope s_0 was found to sometimes fail the χ^2 hypothesis test, and when performing the test explicitly for a specific run the hypothesis had to be rejected, indicating that the slope uncertainties were underestimated. This led us to try an alternative method. Instead of taking the midpoint of each hit cell, we drew a random x -coordinate uniformly from within the fired cell. However, this introduced a more severe problem: the pull widths for all parameters were consistently inflated by a factor of ~ 1.4 . A possible explanation for why the second method performed worse is that the random sampling introduces larger fluctuations, the error on a reconstructed hit position can be up to $\pm \text{cellWidth}$, whereas in the midpoint method the maximum error is only $\pm \text{cellWidth}/2$. Since the problems introduced by the second method were more severe than those of the first, we returned to the midpoint method. The miscalibration of the pull widths in both methods can be explained as a consequence of applying a least-squares fitter, which assumes Gaussian errors, to uniform measurement errors which are inherently non-Gaussian. In the first method, the reason why only the hypothesis for s_0 is sometimes rejected while x_0 is largely accepted stems from how linear regression is calculated. The x_0 estimate depends on the average of all hit positions, so the non-Gaussian errors partially cancel out. The s_0 estimate on the other hand depends on differences between hit positions rather than their average, making it more sensitive to the non-Gaussian errors and therefore more likely to fail the hypothesis test.

The second problem encountered was a systematic bias introduced in the "Momentum Resolution" part of the experiment. When we look more closely at the formula from the appendix,

$$\theta \approx \frac{L}{\rho} = \frac{L \cdot q \cdot B}{p_T}, \quad (11)$$

we see a close resemblance to the arc length relation

$$\theta = \frac{s}{\rho} \iff \theta \cdot \rho = s, \quad (12)$$

where s is the length of the circle segment. This relation is only a good approximation when the bending radius is much larger than the magnet length, i.e. when $L \approx s$.

Upon performing hypothesis tests on the bias, we encountered a rejection rate of roughly 30-40% for the mean of the p_T pull distribution, indicating a significant systematic bias.

To eliminate this bias, we derived an exact expression for p_T .

Starting from equation (12), and treating the circular arc as a sequence of infinitesimally small segments, we write

$$d\theta \cdot \rho = ds. \quad (13)$$

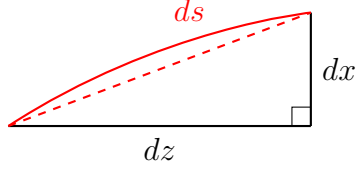


Figure 10: $ds^2 = dz^2 + dx^2$

Each infinitesimal segment can be decomposed along the z and x axes (see Figure 10), given by trigonometry:

$$dz = ds \cdot \cos(\theta), \quad dx = ds \cdot \sin(\theta), \quad (14)$$

where θ is the angle of the particle's tangent with respect to the z -axis. Substituting equation (13) into equation (14):

$$dz = \rho \cdot \cos(\theta) d\theta, \quad dx = \rho \cdot \sin(\theta) d\theta. \quad (15)$$

Integrating the expression for dz from the entry angle θ_{in} to the exit angle θ_{out} yields the magnet length L :

$$L = \int_{\theta_{in}}^{\theta_{out}} \rho \cdot \cos(\theta) d\theta = \rho \cdot (\sin(\theta_{out}) - \sin(\theta_{in})). \quad (16)$$

Combining this with $p_T = q \cdot B \cdot \rho$ gives the exact expression:

$$p_T = \frac{L \cdot q \cdot B}{\sin(\theta_{out}) - \sin(\theta_{in})}. \quad (17)$$

Using this relation reduced the rejection rate to 0 – 1%, indicating that systematic bias has been effectively eliminated.

The third and last problem concerned the interpolation of the curved trajectory through the magnet for visualization purposes.

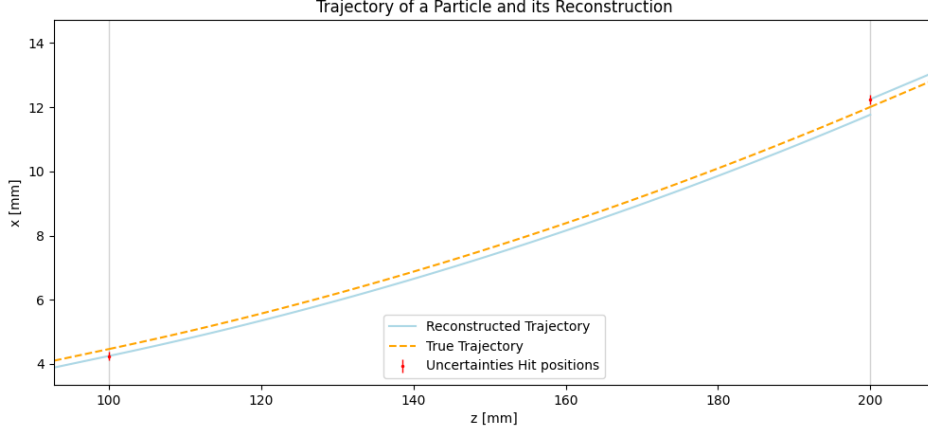


Figure 11: Incorrect interpolation of the reconstructed trajectory through the magnet.

The first approach was to geometrically reconstruct the center of curvature and use the circle equation to obtain x as a function of z . Given the reconstructed entry point (z_{in}, x_{in}) and the entry slope s_{in} , the center of curvature $M = (M_z, M_x)$ lies perpendicular to the tangent at distance ρ :

$$M_z = z_{in} + \rho \sin(\theta_{in}), \quad (18)$$

$$M_x = x_{in} - \rho \cos(\theta_{in}). \quad (19)$$

The trajectory inside the magnet would then follow

$$x(z) = M_x \pm \sqrt{\rho^2 - (z - M_z)^2}. \quad (20)$$

The problem with this approach is that the two independent least-squares fits before and after the magnet each return a slope and intercept determined purely by their respective hits. There is no guarantee that the resulting entry state (z_{in}, x_{in}, s_{in}) and exit state $(z_{out}, x_{out}, s_{out})$ are consistent with a single circle of radius ρ . In general, due to measurement noise in both fits, the constraint

$$(z_{out} - M_z)^2 + (x_{out} - M_x)^2 = \rho^2 \quad (21)$$

will not be exactly satisfied, making a geometrically consistent circular interpolation impossible.

To resolve this, we instead used a cubic Hermite spline to interpolate the trajectory through the magnet. A cubic Hermite spline is uniquely determined by four conditions: the values and first derivatives at both endpoints. These four conditions fix the four coefficients of a cubic polynomial $p(z) = a_0 + a_1z + a_2z^2 + a_3z^3$.

In our case the interpolate satisfies the following:

$$p(z_{in}) = p(100) = x_{in}, \quad p(z_{out}) = p(200) = x_{out}, \quad (22)$$

$$p'(100) = s_{in}, \quad p'(200) = s_{out}, \quad (23)$$

This guarantees C^1 continuity at both magnet boundaries without requiring geometric consistency between the two independent fits. However, it should be noted that a cubic polynomial does not exactly reproduce a circular arc. The interpolated curve only matches the true trajectory at the two boundary points and in its tangent directions there, and will deviate slightly from the physical circular path in between. Since this interpolation is used purely for visualization purposes and not in the momentum reconstruction itself, this geometric inaccuracy has no effect on the physics results.

Appendix

A: Additional plots

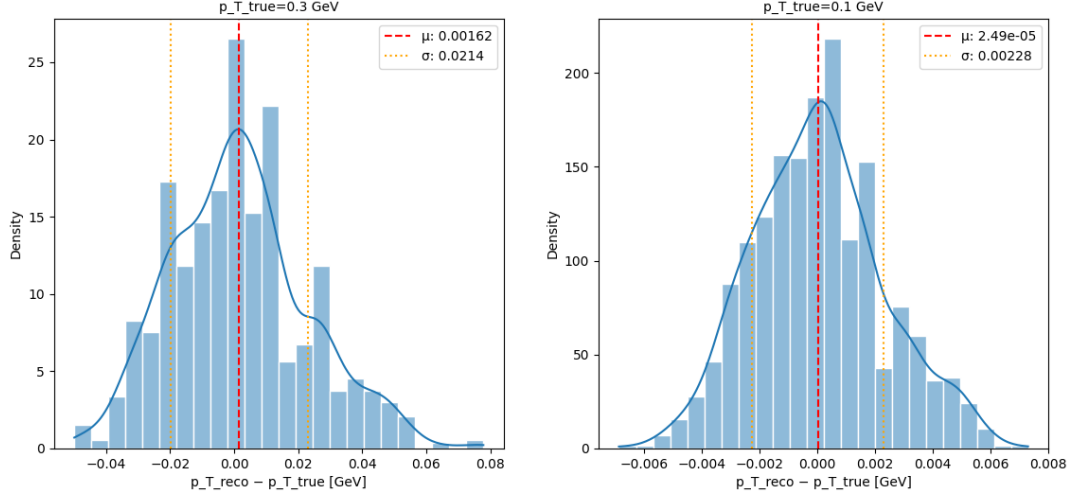


Figure 12: Histograms for p_T^{true} ($\{0.3, 0.1\}$) GeV.

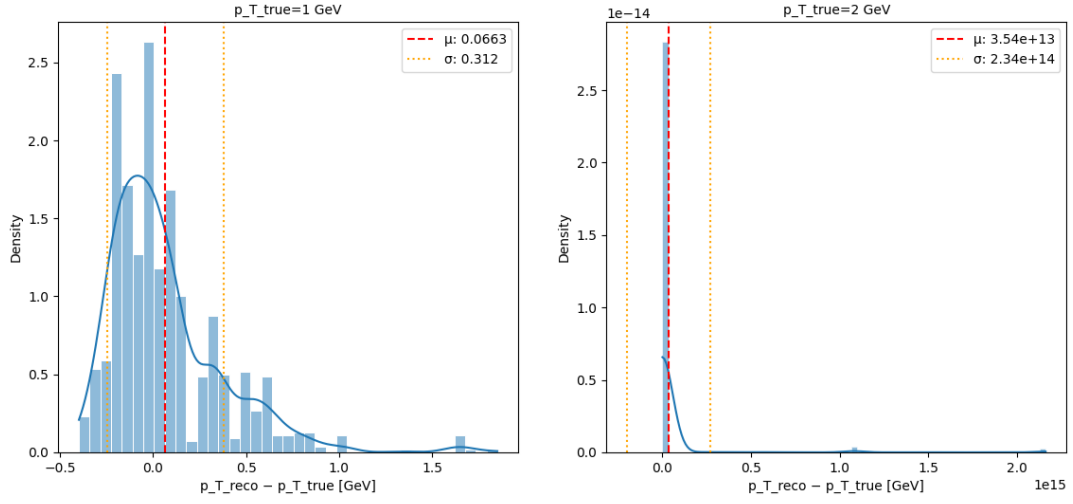


Figure 13: Histograms for p_T^{true} ($\{1, 2\}$) GeV.

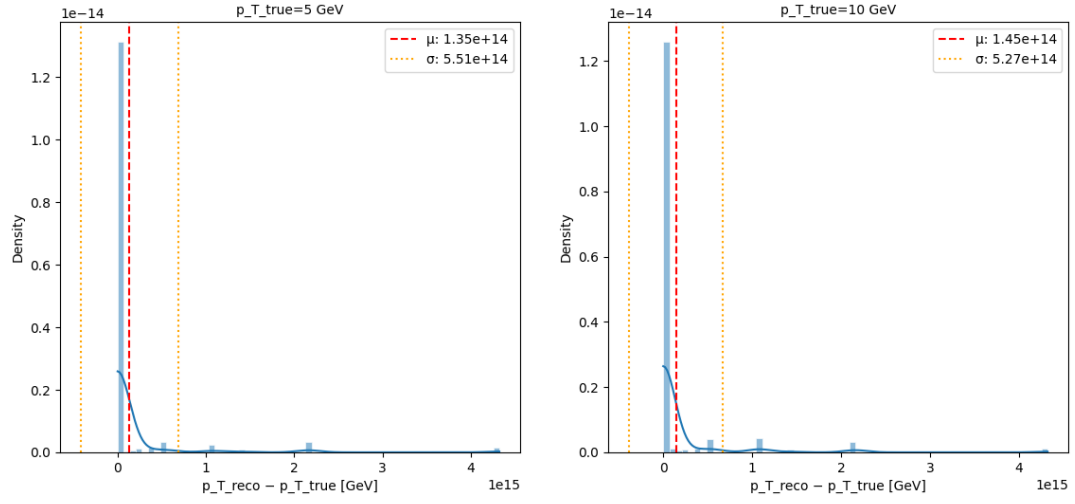


Figure 14: Histograms for p_T^{true} ($\{5, 10\}$) GeV.

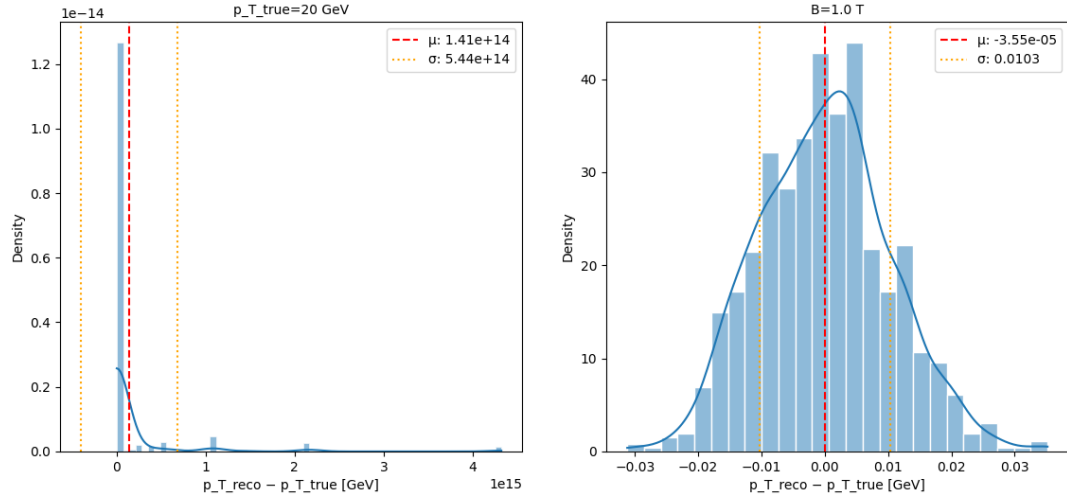


Figure 15: Histograms for $p_T^{\text{true}} = 20$ GeV and $B = 1.0$ T.

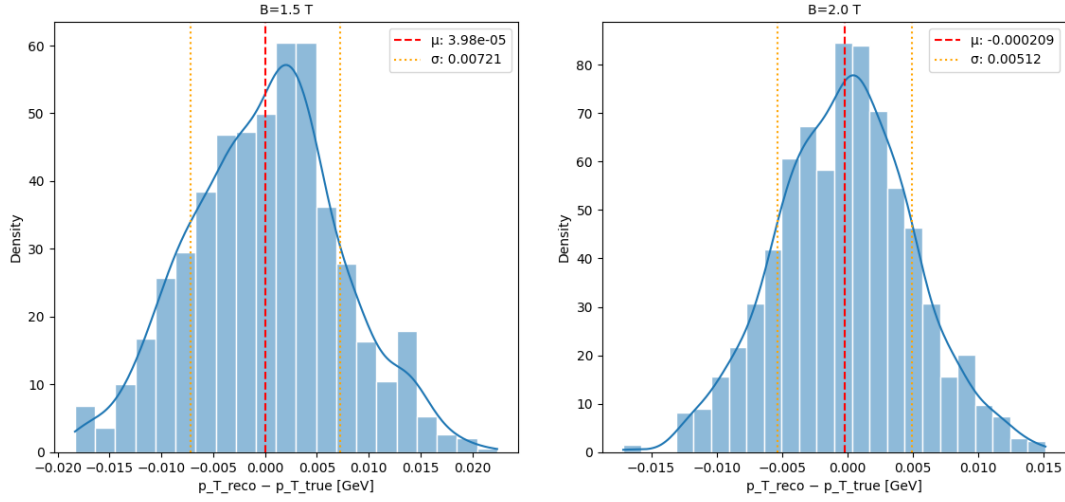


Figure 16: *Histograms for B ($\{1.5, 2.0\}$) T.*

B: Derivation of error propagation formula

Using error propagation, we can obtain the uncertainty of p_T from s_0^{reco} and s_1^{reco} :

$$\sigma(f)^2 = \left| \frac{\partial f}{\partial x} \right|^2 \cdot \sigma_x^2 + \left| \frac{\partial f}{\partial y} \right|^2 \cdot \sigma_y^2 + 2 \cdot \frac{\partial f}{\partial x} \cdot \frac{\partial f}{\partial y} \cdot \text{cov}(x, y) \quad (24)$$

For the first method to reconstruct p_T , we have $p_T(s_0, s_1) = \frac{q \cdot B \cdot L}{|\Delta(s_0, s_1)|}$, where we introduce the shorthand $\Delta(s_0, s_1) = \arctan(s_0) - \arctan(s_1)$

$$\frac{\partial p_T}{\partial s_0}(s_0, s_1) = (q \cdot B \cdot L \cdot |\Delta(s_0, s_1)|^{-1})'_{s_0} \quad (25)$$

$$= q \cdot B \cdot L \cdot \left(\frac{1}{1 + s_0^2} \right) \cdot \left(\frac{-1}{\Delta(s_0, s_1) \cdot |\Delta(s_0, s_1)|} \right) \quad (26)$$

$$\frac{\partial p_T}{\partial s_1}(s_0, s_1) = (q \cdot B \cdot L \cdot |\Delta(s_0, s_1)|^{-1})'_{s_1} \quad (27)$$

$$= q \cdot B \cdot L \cdot \left(\frac{-1}{1 + s_1^2} \right) \cdot \left(\frac{-1}{\Delta(s_0, s_1) \cdot |\Delta(s_0, s_1)|} \right) \quad (28)$$

$$\sigma(p_T) = \sqrt{\left|\frac{\partial p_T}{\partial s_0}\right|^2 \cdot \sigma_{s_0}^2 + \left|\frac{\partial p_T}{\partial s_1}\right|^2 \cdot \sigma_{s_1}^2} \quad (29)$$

$$\begin{aligned} &= \left(\left(q \cdot B \cdot L \cdot \frac{1}{1+s_0^2} \cdot \frac{-1}{\Delta(s_0, s_1) |\Delta(s_0, s_1)|} \right)^2 \cdot \sigma_{s_0}^2 \right. \\ &\quad + \left(q \cdot B \cdot L \cdot \frac{-1}{1+s_1^2} \cdot \frac{-1}{\Delta(s_0, s_1) |\Delta(s_0, s_1)|} \right)^2 \cdot \sigma_{s_1}^2 \\ &\quad \left. + 2 \cdot q \cdot B \cdot L \cdot \frac{-1}{1+s_1^2} \cdot \frac{-1}{\Delta(s_0, s_1) |\Delta(s_0, s_1)|} \cdot \text{cov}(s_0, s_1) \right)^{\frac{1}{2}} \end{aligned} \quad (30)$$

$$\begin{aligned} &= \left((q \cdot B \cdot L)^2 \cdot \frac{\sigma_{s_0}^2}{(1+s_0^2)^2 \cdot |\Delta(s_0, s_1)|^4} \right. \\ &\quad + (q \cdot B \cdot L)^2 \cdot \frac{\sigma_{s_1}^2}{(1+s_1^2)^2 \cdot |\Delta(s_0, s_1)|^4} \\ &\quad \left. + 2 \cdot (q \cdot B \cdot L)^2 \cdot \left(\frac{1}{1+s_0^2} \cdot \frac{-1}{1+s_1^2} \right) \cdot \frac{\text{cov}(s_0, s_1)}{|\Delta(s_0, s_1)|^4} \right)^{\frac{1}{2}} \end{aligned} \quad (31)$$

$$\begin{aligned} &= \left((q \cdot B \cdot L)^2 \cdot \left(\frac{\sigma_{s_0}^2}{(1+s_0^2)^2 \cdot |\Delta(s_0, s_1)|^4} + \frac{\sigma_{s_1}^2}{(1+s_1^2)^2 \cdot |\Delta(s_0, s_1)|^4} \right. \right. \\ &\quad \left. \left. - \frac{2 \cdot \text{cov}(s_0, s_1)}{(1+s_0^2)(1+s_1^2) |\Delta(s_0, s_1)|^4} \right) \right)^{\frac{1}{2}} \end{aligned} \quad (32)$$

Since s_0 and s_1 come from independent least-squares fits, $\text{cov}(s_0, s_1) = 0$. The final uncertainty of p_T is thus:

$$\sigma(p_T) = |q \cdot B \cdot L| \cdot \left(\frac{\sigma_{s_0}^2}{(1+s_0^2)^2 \cdot |\Delta(s_0, s_1)|^4} + \frac{\sigma_{s_1}^2}{(1+s_1^2)^2 \cdot |\Delta(s_0, s_1)|^4} \right)^{\frac{1}{2}} \quad (33)$$

We apply the same procedure to our analytical formula. To reconstruct p_T , we have $p_T = \frac{L \cdot q \cdot B}{\sin(\theta_{\text{out}}) - \sin(\theta_{\text{in}})}$, where $\theta_i = \arctan(s_i)$

$$\frac{\partial p_T}{\partial s_1} = q \cdot B \cdot L \cdot \frac{s_0^2 + 1}{\sqrt{s_1^2 + 1} \left(s_0 \sqrt{s_1^2 + 1} - \sqrt{s_0^2 + 1} s_1 \right)^2} \quad (34)$$

$$\frac{\partial p_T}{\partial s_0} = q \cdot B \cdot L \cdot \frac{s_1^2 + 1}{\sqrt{s_0^2 + 1} \left(s_1 \sqrt{s_0^2 + 1} - \sqrt{s_1^2 + 1} s_0 \right)^2} \quad (35)$$

$$\sigma(p_T)^2 = \left| \frac{\partial p_T}{\partial s_0} \right|^2 \cdot \sigma_{s_0}^2 + \left| \frac{\partial p_T}{\partial s_1} \right|^2 \cdot \sigma_{s_1}^2 \quad (36)$$

$$= \left| \frac{q \cdot B \cdot L \cdot (s_1^2 + 1)}{\sqrt{s_0^2 + 1} \left(s_1 \cdot \sqrt{s_0^2 + 1} - \sqrt{s_1^2 + 1} \cdot s_0 \right)^2} \right|^2 \cdot \sigma_{s_0}^2 + \left| \frac{q \cdot B \cdot L \cdot (s_0^2 + 1)}{\sqrt{s_1^2 + 1} \left(s_0 \cdot \sqrt{s_1^2 + 1} - \sqrt{s_0^2 + 1} \cdot s_1 \right)^2} \right|^2 \cdot \sigma_{s_1}^2 \quad (37)$$

$$= |q \cdot B \cdot L|^2 \cdot \left(\left| \frac{s_0^2 + 1}{\sqrt{s_1^2 + 1} \left(s_0 \cdot \sqrt{s_1^2 + 1} - \sqrt{s_0^2 + 1} \cdot s_1 \right)^2} \right|^2 \cdot \sigma_{s_1}^2 + \left| \frac{s_1^2 + 1}{\sqrt{s_0^2 + 1} \left(s_1 \cdot \sqrt{s_0^2 + 1} - \sqrt{s_1^2 + 1} \cdot s_0 \right)^2} \right|^2 \cdot \sigma_{s_0}^2 \right) \quad (38)$$

$$= |q \cdot B \cdot L|^2 \left(\frac{(s_0^2 + 1)^3 \cdot \sigma_{s_1}^2}{(s_0^2 + 1)(s_1^2 + 1) \left(s_0 \cdot \sqrt{s_1^2 + 1} - \sqrt{s_0^2 + 1} \cdot s_1 \right)^4} + \frac{(s_1^2 + 1)^3 \cdot \sigma_{s_0}^2}{(s_1^2 + 1)(s_0^2 + 1) \left(s_0 \cdot \sqrt{s_1^2 + 1} - \sqrt{s_0^2 + 1} \cdot s_1 \right)^4} \right) \quad (39)$$

$$= |q \cdot B \cdot L|^2 \left(\frac{(s_0^2 + 1)^3 \cdot \sigma_{s_1}^2 + (s_1^2 + 1)^3 \cdot \sigma_{s_0}^2}{(s_1^2 + 1)(s_0^2 + 1) \left(s_0 \cdot \sqrt{s_1^2 + 1} - \sqrt{s_0^2 + 1} \cdot s_1 \right)^4} \right) \quad (40)$$

Again, because the two least-squares fits are independent, the covariance term vanishes and we obtain:

$$\sigma(p_T) = |q \cdot B \cdot L| \cdot \frac{\sqrt{(s_0^2 + 1)^3 \sigma_{s_1}^2 + (s_1^2 + 1)^3 \cdot \sigma_{s_0}^2}}{\sqrt{s_1^2 + 1} \sqrt{s_0^2 + 1} \left(s_0 \sqrt{s_1^2 + 1} - s_1 \sqrt{s_0^2 + 1} \right)^2} \quad (41)$$